

Paper Title: Forecasting X with Y — A Comparative Evaluation

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Abstract

Liquidity management ensures that a bank has sufficient cash to fulfill its short-term obligations, while minimizing the cost of holding an excess of it. One of the most important sources of liquidity for a bank are non-maturity deposits. Therefore, forecasting the volume of these deposits is crucial for the liquidity management process of a bank. However, despite the economic importance of deposit volume prediction, the scientific research is limited, meaning there is not an established scientific framework to correctly model them. Here, we build different machine learning models for deposit volume prediction and evaluate whether model-robustness provides more accurate results. + (Maybe compare with Elias models) ++ Main results?. We expect that this study contributes to the development of more specialized approaches for modeling non-maturity deposits, while also support future research on other practical banking applications besides liquidity management, including strategic planning, interest rate risk management, and profitability analysis. Placeholder abstract text (target: ≤ 150 words, no citations).

Contents

1	Introduction	3
2	Related Work	4
3	Data and Preprocessing	4
3.1	Target Variable and Explanatory Variables	4
3.2	Data Alignment and Transformation: Interpolation Methods	5
4	Forecasting Models	5
4.1	Forecasting Problem	5
4.2	Forecasting Models	7
4.3	Covariate Specifications	8
4.4	Rolling-Origin Evaluation	9
4.5	Evaluation	9
5	Results	9
5.1	Forecasting Performance	9
5.2	Effect of Covariate Selection	10
5.3	Variation Across Forecast Origins	10
6	Conclusion and Outlook	10
A	Appendix: Additional Figures and Tables	11
B	Appendix: Code	11

1 Introduction

The Basel Committee on Banking Supervision (BCBS, 2016) defines non-maturity deposits (NMDs) as liabilities without a contractual maturity where the depositor is free to withdraw at any time. They consist mainly of fully transferable deposits and non-transferable deposits that can be converted by end of the next business day. The interest paid from NMDs is significantly below other products issued by the banks. Moreover, since there is no fixed maturity date, banks reserve the right to change this interest rate at any time. Therefore, because of their stability and low-cost, NMDs are an important source of funding for banks. Especially for German banks, NMDs nowadays represent the most important funding source after exhibiting a dynamic growth in the aftermath of the financial crisis of 2008 (Schlueter et al., 2015). Considering this information, it becomes clear how important is the study of NMDs deposits for banks liquidity management, the process of assessing the liquidity needs of a bank and supplying or absorbing the appropriate amount of liquidity through open market operations (European Central Bank, 2011). However, because of their stochastic nature, the study of these products become, to say the least, extremely challenging. Despite its importance, the academic research on this topic is still scarce. Recent years have seen a growing interest on the subject in the form of graduate theses, journals and papers (Winkler, 2025; Marena et al., 2022). Nevertheless, this current literature presents a wide range of approaches, from economic to statistical and mathematical models, there is yet to be a general accepted scientific consensus to the optimal methodology for modeling NMDs. Hence, the need for a structured and effective methodology for the forecasting of these financial products is evident.

This study addresses the identified research gap by developing a comparative framework for forecasting non-maturity deposit volumes. We evaluate naive benchmarks, statistical models, and machine-learning models across alternative covariate sets and multiple forecast origins representing different economic environments. Although the study is motivated by NMD modelling, the empirical analysis uses publicly available German household overnight deposit data as a proxy for aggregate household NMD volumes.

The study makes three contributions. First, it examines whether greater model complexity leads to more accurate forecasts. Second, it evaluates how interest-rate, macroeconomic, and financial-market covariates affect forecasting performance. Third, it investigates whether model performance remains stable across different economic periods. All models are assessed using a common rolling-origin design and a realistic information set in which realized economic covariates beyond the forecast origin are unavailable.

The remainder of this paper is structured as follows. Section 2 provides related literature on NMD modeling and forecasting, and positions this study within this research. Section 3 presents the data and forecasting methodology, including data preparation, model explanation, covariates, and rolling-origin scheme. Section 4 reports and compares the results. Finally Section 6 concludes the paper.

Research Question(s):

- RQ1** How accurately can statistical and machine-learning models forecast German household overnight deposit volumes over a 100-day horizon?
- RQ2** Do nonlinear machine-learning models outperform linear and naive benchmark models under the same rolling-origin evaluation framework?
- RQ3** Which groups of covariates provide the most improvement in forecasting performance?

2 Related Work

Table 1: Overview of related work and positioning of this paper.

Reference	Data/Market	Method	Rolling	Out-of-sample	...
Author et al. (Year)	...	...	✗	✓	...
Author et al. (Year)	...	...	✓	✗	...
This paper	...	...	✓	✓	...

3 Data and Preprocessing

3.1 Target Variable and Explanatory Variables

Detailed non-maturity deposits (NMDs) data are generally not publicly available because it reveals sensitive bank information. For this study, we utilize public end-of-period levels of overnight deposits held by private households data, published by the Deutsche Bundesbank (Deutsche Bundesbank, 2025a) as a proxy for aggregate NMD volumes. Household overnight deposits provide an accurate publicly observable proxy since they are a subset of NMDs and account for a substantial share of the banks’ total deposit funding (Deutsche Bundesbank, 2024). Therefore, the aggregate volume of household overnight deposits are likely to respond to several of the same economic drivers. The dataset covers the period January 2003 to September 2025 and the series is measured in billions of euros.

Euro Short-Term Rate (€STR): sourced from the ECB (European Central Bank, 2025b), is available at daily frequency and captures the average overnight borrowing rate of banks in the euro area. As each published value reflects the previous business day’s transactions, it constitutes a direct, transaction-based measure of interbank market conditions. While €STR was officially introduced in October 2019, the sample period extends back to January 2003. For the period March 2017 to September 2019, the Pre-€STR (European Central Bank, 2025a) is used as a direct predecessor series. For earlier observations, the series is reconstructed following Winkler (2025, Ch. 5.2). The reasoning for including €STR as a covariate is that a shift from it could cause a shift in the rates banks apply to deposit accounts, meaning a change in the total overnight deposits.

German geopolitical risk index (GPRC_DEU): published on a monthly basis (Caldara and Iacoviello, 2022), quantifies the geopolitical risk in Germany by using automated text search in the U.S. newspapers *New York Times*, *Chicago Tribune* and *Washington Post*. Geopolitical context may have an influence because as uncertainty rises, customers might be more drawn to postpone spending and leaving cash in their accounts, leading to an increase in overnight deposits.

Month-on-month inflation (MoM-Inflation): sourced from Destatis (Destatis, 2025), calculates the percentage change in the Consumer Price Index relative to the preceding month. Inflation changes how much households spend and save, therefore it could also lead to explain movements of their total overnight deposits.

DAX daily closing prices: obtained via the yfinance Python library (Aroussi, 2025), retrieving data from Yahoo Finance (Yahoo Finance, 2026) for the ticker `^GDAXI`. As DAX reflects the changes in the German stock market as well as investor confidence, this may influence the total amount hold by private households.

10-year German government bond yield (10Y Bond): sourced at business-day frequency from the Deutsche Bundesbank (Deutsche Bundesbank, 2025b). As the 10-year old government bond is a relatively safe investment product, private household may be drawn to invest more on them if the rate is sufficiently attractive, investing less in overnight deposits account.

Einlagezinssatz: it refers to the deposit rate. The more interest households receive on deposits, the more attractive it may be to keep their money there.

3.2 Data Alignment and Transformation: Interpolation Methods

Since multiple variables are published at monthly frequency, they therefore require interpolation to a daily grid before model training. Following Winkler (2025, Ch. 5.2), piecewise cubic Hermite interpolating polynomial (PCHIP) interpolation—introduced by Fritsch and Carlson (1980)—outperforms than linear and autoregressive methods for financial time series. PCHIP interpolation is applied to GPRC_DEU and MoM Inflation, preserving local monotonicity of the input series and avoiding artificial overshoots between monthly observations (Fritsch and Carlson, 1980). The Einlagevolumen and Einlagezinssatz series, as well as the three-period €STR reconstruction described above, are constructed following the methodology of Winkler (2025, Ch. 5.1–5.2). The DAX closing prices and the 10-year bond yield are natively available at daily frequency and require no interpolation. For variables available at business-day frequency (€STR and 10Y Bond), weekend and public holiday gaps are handled via forward-fill, carrying the last observed value forward to the following business day. This ensures a complete daily index across the full sample period while avoiding the introduction of artificial variation on non-trading days. The following figure illustrates all variables over the full sample period after interpolation to daily frequency. As evident from Figure 1, the Einlagevolumen exhibits a clear upward trend throughout the timeline, indicating nonstationarity in the mean. To this first differencing is applied, to obtain a stationary target series and to limit accumulating directional errors (Box et al., 2015), as illustrated in Figure 2. Second differencing was rejected do to over-differencing.

After interpolation to daily frequency, the dataset comprises 1517 observations covering the full sample period January 2003 to December 2025.

4 Forecasting Models

4.1 Forecasting Problem

Let y_t denote the volume of German household overnight deposits at time t , and let $\mathbf{x}_t$ denote the vector of macroeconomic and financial covariates. The objective is to forecast deposit volumes over a 100-day horizon using the information available during the preceding 30 days.

Because the deposit-volume series exhibits a strong long-term trend, the models are trained to predict its first difference,

$$\Delta y_t = y_t - y_{t-1}. \quad (1)$$

At each forecast origin t , the forecasting models use a lookback window of length $L = 30$. The information set therefore contains the historical deposit volume and the selected covariates over the period $t - L + 1, \dots, t$. Each model produces the complete vector of predicted changes for the following $H = 100$ days:

$$\widehat{\Delta \mathbf{y}}_{t+1:t+H} = f(y_{t-L+1:t}, \mathbf{x}_{t-L+1:t}), \quad L = 30, \quad H = 100. \quad (2)$$

The study therefore applies a direct multi-output forecasting strategy. Rather than generating forecasts recursively one day at a time, each model estimates all 100 future changes simultaneously. The predicted changes are subsequently converted back into deposit-volume levels according to

$$\hat{y}_{t+h} = y_t + \sum_{j=1}^h \widehat{\Delta y}_{t+j}, \quad h = 1, \dots, H.$$

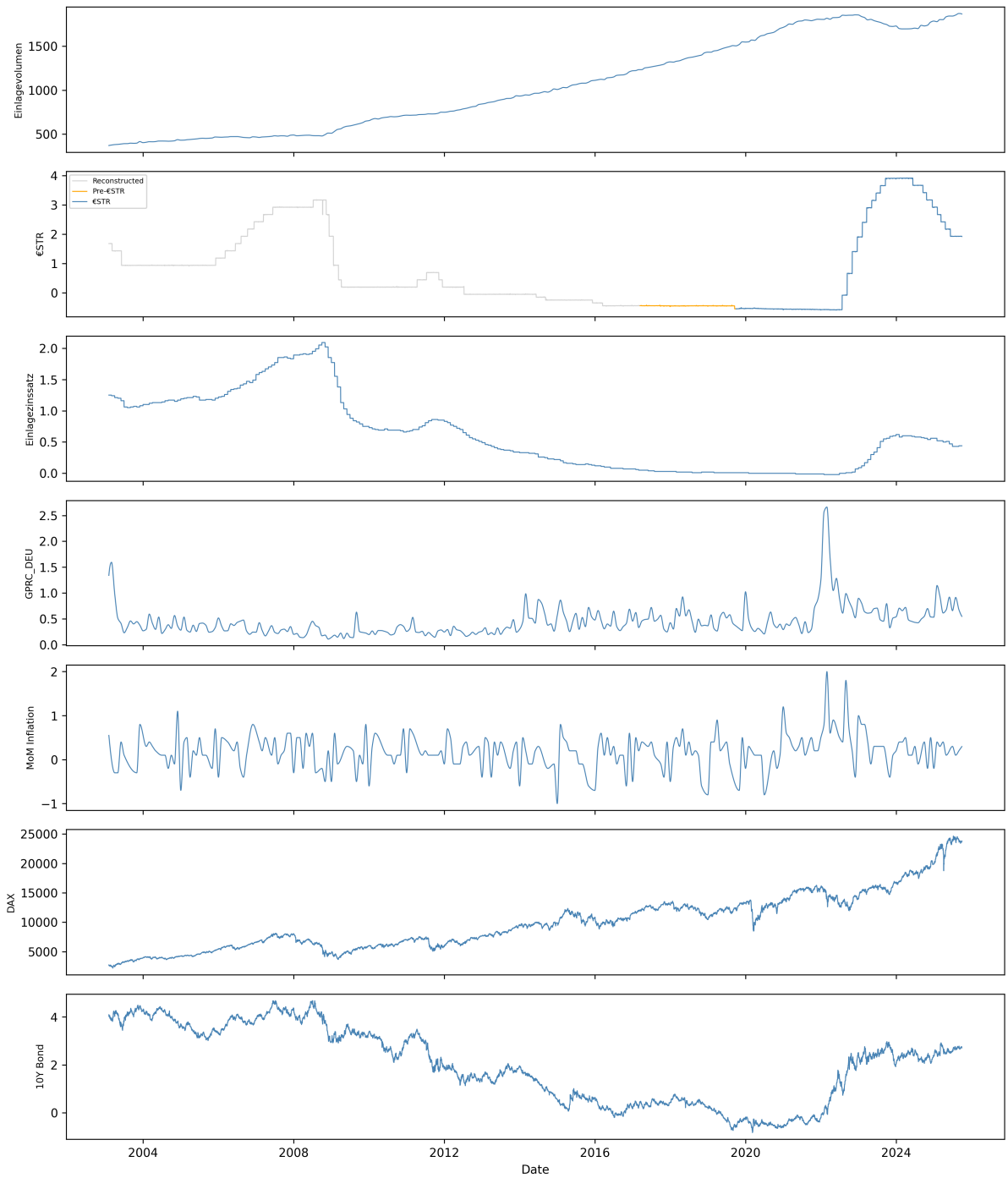


Figure 1: Time series of all variables over the sample period January 2003 – December 2025. All monthly series are shown after PCHIP interpolation to daily frequency. The €STR series begins in October 2019 following its introduction by the ECB.

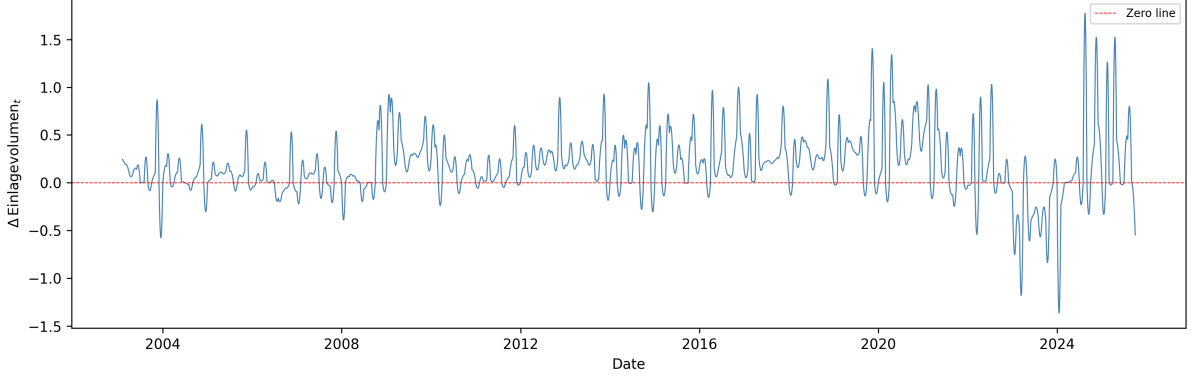


Figure 2: First difference of Einlagevolumen_t over the full sample period January 2003 – December 2025. The series fluctuates around zero throughout, confirming stationarity after differencing. Increased volatility from 2022 onward coincides with the ECB interest rate tightening cycle.

This reconstruction permits the models to be trained on the less strongly trending differenced series while their performance is evaluated in terms of the economically interpretable deposit-volume level.

At each forecast origin, only observations available up to that date are used for model estimation and feature scaling. Realized future values of the economic covariates are not supplied to the models. Calendar variables whose future values are known in advance, such as the day of the week or month-end indicator, may be included where required by the model architecture. The model-specific treatment of future covariates is describe later.

4.2 Forecasting Models

Naive benchmarks. Random Walk extrapolates the last observed daily change at a constant rate,

$$\hat{y}_{t_0+h} = y_{t_0} + h \cdot \Delta y_{t_0}, \quad h = 1, \dots, 100,$$

which corresponds to an $\text{ARIMA}(0, 1, 0)$ process with drift (Box et al., 2015, Ch. 4.1.2). Naive Hold instead assumes the level remains constant, $\hat{y}_{t_0+h} = y_{t_0}$. Both benchmarks are fully interpretable, require no covariates, serving as a lower bound against which the higher complexity of the remaining models must be justified.

Linear Regression (Ridge). A multi-output ridge regression maps the flattened lookback window (target and covariates, optionally including lags) to the full 100-step

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forecast vector directly (a “direct” rather than “iterated” multi-step strategy), regularized by an ℓ_2 penalty α selected via time-series cross-validation (Hastie et al., 2009, Ch. 3). As a linear model, Ridge retains full interpretability of covariate effects through its coefficients, at the cost of being unable to capture non-linear relationships between covariates and deposit behavior.

XGBoost. Gradient-boosted regression trees are trained on the same flattened input representation, with hyperparameters (max depth, learning rate, subsample ratio) selected via grid search under `TimeSeriesSplit` cross-validation to avoid look-ahead bias in model selection (Chen and Guestrin, 2016). Boosted trees can capture non-linear covariate interactions that Ridge cannot (Hastie et al., 2009, Ch. 10), while remaining moderately interpretable through feature importance measures – positioning XGBoost as an intermediate step between the fully interpretable linear model and the black-box neural architectures that follow.

Feedforward Neural Network (FFN). A fully connected network with two hidden layers (128, 64 units, ReLU activation) is trained to directly predict the 100-step output vector. Following the general formulation of a feedforward layer (Hastie et al., 2009, Ch. 11) (cf. also Winkler (2025, Ch. 4.2)),

$$z_j = \sum_{i=1}^n w_{ji}x_i + b_j, \quad a_j = \phi(z_j),$$

where ϕ is the ReLU activation for hidden layers. Training uses Adam with early stopping on a held-out validation slice (last 15% of the training window, chronologically), monitoring validation MAE with a patience of 20 iterations. Unlike Ridge and XGBoost, the FFN offers no direct interpretability of individual covariate effects, trading transparency for greater representational flexibility.

Temporal Fusion Transformer (TFT). Unlike the sequence models above, the TFT explicitly distinguishes between *known* future inputs (deterministically known for the entire forecast horizon – here, calendar features: weekday, month-end indicator, day-of-month) and *unknown* future inputs (the six macro-financial covariates, whose future values are not known and are therefore held at their last observed value throughout the forecast horizon, consistent with the realistic no-future-leakage scenario described in Section ??) (Lim et al., 2021). The model is trained with early stopping (patience 5 epochs) on validation loss. Despite being the most complex model considered, the TFT’s attention mechanism was explicitly designed by Lim et al. (2021) to offer a degree of interpretability – e.g. via variable selection weights – that purely black-box architectures typically lack, distinguishing it from a conventional deep learning model of comparable complexity.

Ensembling. Each sequence model is fit S times with different random seeds (and, when $S > 1$, bootstrap-resampled training data) to form an ensemble; point forecasts are the ensemble mean, and 90%/98% confidence bands are the corresponding empirical percentiles (5/95 and 1/99) across ensemble paths, following the same visualization convention as Winkler (2025, Ch. 5.4).

4.3 Covariate Specifications

To assess the predictive value of different economic information, each non-naive model is evaluated using the nine predefined covariate sets shown in Table 2. These specifications compare economically related variable groups, individual interest-rate variables, and ablations of the complete information set. The naive benchmarks use no covariates.

Table 2: Covariate specifications

Specification	Included variables
All	€STR, deposit rate, geopolitical risk, inflation, DAX, and ten-year German government bond yield
Interest rates	€STR and deposit rate
Financial markets	DAX and ten-year government bond yield
Macroeconomic	Geopolitical risk and inflation
Rates and macroeconomy	€STR, deposit rate, geopolitical risk, and inflation
Excluding DAX	All variables except DAX
Excluding inflation	All variables except inflation
€STR only	€STR
Deposit rate only	Deposit rate

4.4 Rolling-Origin Evaluation

Forecasting performance is assessed using an expanding-window rolling-origin design. At each forecast origin o_k , all observations available up to that date form the training sample, while the following $H = 100$ days form the evaluation window:

$$\mathcal{T}_k = \{1, \dots, o_k\}, \quad \mathcal{E}_k = \{o_k + 1, \dots, o_k + H\}. \quad (3)$$

The models are re-estimated at every origin as newly observed data become available. The evaluation uses four origins: 1 January 2009, 1 January 2013, 1 March 2019, and 1 June 2025.

All preprocessing parameters, including feature scaling, are estimated using the corresponding training sample only. Realized economic covariates beyond the forecast origin are not supplied to the models. For the Temporal Fusion Transformer, future calendar variables are known in advance, whereas economic covariates are held at their latest observed values throughout the forecast horizon.

4.5 Evaluation

Performance is evaluated for both the predicted first differences and the reconstructed deposit-volume levels. Let $y_{k,h}$ and $\hat{y}_{k,h}$ denote the observed and predicted volume at horizon h after origin o_k . The evaluation metrics are

$$\begin{aligned} \text{MAE}_k &= \frac{1}{H} \sum_{h=1}^H |y_{k,h} - \hat{y}_{k,h}|, \\ \text{RMSE}_k &= \sqrt{\frac{1}{H} \sum_{h=1}^H (y_{k,h} - \hat{y}_{k,h})^2}, \\ R_k^2 &= 1 - \frac{\sum_{h=1}^H (y_{k,h} - \hat{y}_{k,h})^2}{\sum_{h=1}^H (y_{k,h} - \bar{y}_k)^2}, \end{aligned}$$

where $\bar{y}_k$ is the mean observed volume in evaluation window k . Negative R^2 values indicate performance below the test-window mean benchmark.

Metrics are calculated separately at each forecast origin and summarized by their mean and standard deviation across origins. The primary comparison uses MAE for the reconstructed deposit-volume levels, measured in billions of euros. RMSE and R^2 provide complementary information, while metrics on first differences are used as diagnostic measures.

Figure 3: Overview of the analytics pipeline: data preprocessing, model training, and evaluation stages.

Figure 4: Example caption style: “Comparison of forecasting accuracy in terms of RMSE, measured from [start date] to [end date] across [list of models].”

5 Results

5.1 Forecasting Performance

Table 3 reports the best-performing covariate specification for each model according to mean level MAE. The results are descriptive because the covariate sets are selected using the same four evaluation windows on which performance is reported.

Table 3: Forecasting performance across four rolling origins. MAE and RMSE are measured in EUR billions and reported as mean $\pm$ standard deviation.

Model	Best covariate set	MAE	RMSE	Mean R^2
Linear model	All covariates	9.36 ± 11.11	10.26 ± 12.08	0.178
TFT	Macroeconomic	10.78 ± 12.61	12.71 ± 14.18	-0.361
Last-change extrapolation	None	11.78 ± 12.81	14.41 ± 14.25	-0.564
FFN	Macroeconomic	11.85 ± 6.90	15.22 ± 10.35	-0.605
XGBoost	Excluding inflation	12.71 ± 13.28	14.90 ± 14.79	-0.784
Constant-level benchmark	None	17.38 ± 10.13	21.03 ± 10.52	-2.246

The linear model achieved the lowest average MAE and was the only approach with a positive mean R^2 . TFT ranked second, followed by the last-change benchmark. The FFN and XGBoost did not improve upon this simple benchmark on average, while the constant-level forecast produced the largest errors. Therefore, greater model complexity did not consistently lead to higher forecasting accuracy.

5.2 Effect of Covariate Selection

The preferred covariate specification differed across models. The linear model performed best with all six explanatory variables, whereas the FFN and TFT obtained their lowest MAE using only geopolitical risk and inflation. XGBoost performed best when inflation was excluded. These findings suggest that the predictive value of a covariate depends on how the model represents linear and nonlinear relationships. They should not, however, be interpreted as evidence of causal effects.

5.3 Variation Across Forecast Origins

Forecasting performance varied substantially across the four evaluation periods, as reflected by the large standard deviations in Table 3. The FFN displayed the lowest variation in MAE, but its average error remained above those of the linear model and TFT. The results indicate that no model dominated at every forecast origin and that model rankings depend on the economic environment.

Because the evaluation contains only four forecast origins, the results should be interpreted as a comparison across selected historical periods rather than as evidence of statistical superiority. Nevertheless, the benchmark shows that simple and interpretable methods remain competitive for forecasting aggregate household overnight deposits.

6 Conclusion and Outlook

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A Appendix: Additional Figures and Tables

B Appendix: Code

Listing 1: Example: model fitting excerpt

```

1 def fit_sequence_model(model_name, X_train, y_train, seed=SEED):
2     # ...
3     pass

```